

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1. Hospital Patient Record System

A hospital system compares every patient record with every other record to detect duplicate entries using nested loops. Recall what nested loops imply in complexity analysis, explain why this approach increases execution time, and determine both the time complexity and space complexity using asymptotic notation.

Q2. Dictionary Search System

A digital dictionary system uses divide-and-conquer to search entries. Explain how input size reduces at each step. Determine the time complexity of the algorithm. Analyze its efficiency for large datasets. Interpret the result using asymptotic notation.

Code of Conduct

I Venkata Vasu Deva Reddy (192424182) _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Venkata Vasu Deva Reddy

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. Hospital Patient Record System

A hospital system compares every patient record with every other record to detect duplicate entries using nested loops. Recall what nested loops imply in complexity analysis, explain why this approach increases execution time, and determine both the time complexity and space complexity using asymptotic notation.

Hospital Patient Record System

Problem Statement

A hospital system compares every patient record with every other record to detect duplicate entries using **nested loops**.

What are Nested Loops?

Nested loops are loops in which one loop is placed inside another loop. The inner loop executes completely for every iteration of the outer loop.

For example:

```
for(i = 1 to n)
  for(j = 1 to n)
    compare records
```

Here, each patient record is compared with every other patient record.

Why Does This Increase Execution Time?

- Suppose there are **n** patient records.
- The outer loop runs **n** times.
- For each iteration of the outer loop, the inner loop also runs **n** times.
- Therefore, the total number of comparisons is:

$$n \times n = n^2$$

As the number of patient records increases, the number of comparisons grows very rapidly, making the algorithm slower for large datasets.

Time Complexity Using Asymptotic Notation

The total number of operations is:

$$n \times n = n^2$$

Therefore,

- **Big-O (Worst Case):**

$$O(n^2)$$

- **Big-Theta (Exact Complexity):**

$$\Theta(n^2)$$

- **Big-Omega (Best Case):**

$$\Omega(n^2)$$

Hence,

$$T(n) = \Theta(n^2)$$

Space Complexity

The algorithm uses only a few variables (such as loop counters) and does not require any additional data structures.

Therefore,

$$\Theta(1)$$

(Constant space)

Performance Analysis for Large Hospital Datasets

- Every patient record is compared with every other patient record.
- For **1,000** records:

$$1000 \times 1000 = 1,000,000$$

comparisons.

- For **10,000** records:

$$10,000 \times 10,000 = 100,000,000$$

comparisons.

Thus, the execution time increases **quadratically**, making the approach inefficient for very large hospital databases.

Interpretation Using Asymptotic Notation

Asymptotic notation describes how an algorithm grows as the input size increases.

- **Big-O:** $O(n^2)$ – Worst-case time complexity.
- **Big-Theta:** $\Theta(n^2)$ – Exact growth rate.
- **Big-Omega:** $\Omega(n^2)$ – Best-case lower bound.
- **Space Complexity:** $\Theta(1)$ – Constant extra memory.

Final Answer

- **Approach:** Nested loops compare every patient record with every other record.
- **Master Theorem:** **Not Applicable**, because there is **no recurrence relation**.
- **Time Complexity:** $\Theta(n^2)$
- **Space Complexity:** $\Theta(1)$
- **Asymptotic Notation:** $O(n^2) = \Theta(n^2) = \Omega(n^2)$
- **Performance:** Execution time grows quadratically, making this method less efficient for very large hospital datasets.

• Q2. Dictionary Search System

A digital dictionary system uses divide-and-conquer to search entries. Explain how input size reduces at each step. Determine the time complexity of the algorithm. Analyze its efficiency for large datasets. Interpret the result using asymptotic notation.

Dictionary Search System

Problem Statement

A digital dictionary system uses the **divide-and-conquer** approach to search for a word (similar to **Binary Search**).

How Input Size Reduces at Each Step

In a divide-and-conquer algorithm, the search space is divided into **two equal halves** after every comparison.

For example:

- Initial dictionary entries = **1024**
- 1st step → 512 entries
- 2nd step → 256 entries
- 3rd step → 128 entries
- ...
- Continue until only one entry remains.

Thus, the input size reduces as:

$$n, \frac{n}{2}, \frac{n}{4}, \frac{n}{8}, \dots, 1$$

Since the input size is halved at every step, the number of recursive calls is:

$$\boxed{\log_2 n}$$

Step 1: Form the Recurrence Relation

Binary Search follows the recurrence:

$$T(n) = T\left(\frac{n}{2}\right) + 1$$

where:

- $a = 1$
- $b = 2$
- $f(n) = 1$

Step 2: Apply the Master Theorem

The standard form is:

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

Here,

- $a = 1$
- $b = 2$
- $f(n) = 1$

Compute

$$n^{\log_b a} = n^{\log_2 1} = n^0 = 1$$

Since

$$f(n) = \Theta(1) = \Theta(n^0),$$

this satisfies **Case 2** of the Master Theorem.

According to Case 2,

$$T(n) = \Theta(n^0 \log n) = \Theta(\log n)$$

Time Complexity

$$\boxed{\Theta(\log n)}$$

Space Complexity

The recursion depth is

$$\boxed{\Theta(\log n)}$$

If an **iterative binary search** is used, the space complexity becomes

$$\boxed{\Theta(1)}$$

Efficiency for Large Datasets

- The algorithm eliminates **half of the dictionary entries** in every step.
- Even if the dictionary contains **1 million words**, only about

$$\log_2(1,000,000) \approx 20$$

comparisons are required.

Therefore, the divide-and-conquer approach is **very efficient** for large datasets because the execution time increases very slowly as the number of entries grows.

Interpretation Using Asymptotic Notation

Asymptotic notation describes the growth rate of an algorithm.

- **Big-O (Worst Case):**

$$\boxed{O(\log n)}$$

- **Big-Theta (Exact Complexity):**

$$\Theta(\log n)$$

- **Big-Omega (Best Case):**

$$\Omega(\log n)$$

Thus,

$$O(\log n) = \Theta(\log n) = \Omega(\log n)$$

Final Answer

- **Algorithm Used:** Divide-and-Conquer (Binary Search)
- **Recurrence Relation:** $T(n) = T(n/2) + 1$
- **Parameters:** $a = 1, b = 2, f(n) = 1$
- **Master Theorem Case:** Case 2
- **Time Complexity:** $\Theta(\log n)$
- **Space Complexity:** $\Theta(\log n)$ (recursive) or $\Theta(1)$ (iterative)
- **Performance:** The input size is halved at every step, making the algorithm highly efficient for searching very large dictionary datasets.
- **Asymptotic Notation:** $O(\log n) = \Theta(\log n) = \Omega(\log n)$